

# Zehua Wang

github.com/zehuaw1

Email: zehuaw@mit.edu

Website: zehuaw1.github.io

## EDUCATION

- **Massachusetts Institute of Technology — GPA: 5.00/5.00** Cambridge, MA  
*B.S. Candidate in Physics and Artificial Intelligence and Decision Making* Expected May 2028
- **Tsinghua University — GPA: 3.95/4.00** Beijing, China  
*Preparatory Program and First-Year Coursework, Institute for Interdisciplinary Information Sciences (IIIS) Aug 2024 – Jul 2025*

## HONORS & AWARDS

- **Gold Medal (1st in Theory)**, 54th International Physics Olympiad (IPhO) Jul 2024
- **1st Place**, MIT Informatics Tournament (MITIT) 2025 Winter Contest Beginner's Round Dec 2025
- **2nd Place, Modal Prize**, HackMIT 2025 Sep 2025

## EXPERIENCE

- **Sophomore Intern, Algorithm Development & Software Engineering** New York, NY  
*Hudson River Trading* May 2026 – Aug 2026
  - Earned **return offers from both Algorithm Development and Software Engineering** following a sophomore-year internship; plan to pursue the full-time Algo Developer offer.
  - Built and live-tested a trading strategy in Algo Dev; improved existing internal tooling and built new tools in SWE.
- **Undergraduate Researcher, Diffusion Models & Molecular Dynamics** Cambridge, MA  
*MIT — Advisor: Tommi Jaakkola* Feb 2026 – May 2026
  - Co-developed Diffusion Accelerants, a non-Markovian biasing method that augments molecular dynamics with learned measure transport.
- **Undergraduate Researcher, Reinforcement Learning & Embodied AI** Cambridge, MA  
*FortyFive Labs — Advisor: Ge Yang* Oct 2025 – Apr 2026
  - Developed and deployed a functional-compliance whole-body control policy with explicit force estimation and directional, time-varying stiffness; ran RL and imitation-learning experiments across nearly 100 NVIDIA RTX PRO 6000 GPUs and iterated sim-to-real on Unitree G1.
- **Undergraduate Researcher, Learning-Based Control** Beijing, China  
*Tsinghua University — Advisor: Huazhe Xu* Mar 2025 – Aug 2025
  - Developed a multistage RIR (RL-to-imitation-to-real-world) pipeline for Franka manipulation, combining PPO/DrQ-v2 specialists, multitask imitation learning, and sim-to-real transfer.

## PUBLICATIONS

- **Learning Functional Compliance in Whole-body Manipulation**  
Gio Huh\*, Yue Zhuo\*, **Zehua Wang\***, Alan Yu, Yibo Cheng, Yanbing Han, Gabriel B. Margolis, Ge Yang.  
*Conference on Robot Learning (CoRL)*, 2026. [OpenReview] \* Equal contribution.
- **Diffusion Accelerants: Towards Augmenting Molecular Dynamics with Learned Measure Transport**  
Bowen Jing, **Zehua Wang**, Paul Gutkovich, Peter Holderrieth, Tommi Jaakkola.  
*SPIGM Workshop at ICML 2026* (poster). [Paper]
- **Family-Aware Residual Architecture for Predicting Quantum Circuit Simulation Performance**  
Honjar Xing\*, Yehong Jiang\*, Xianbang Wang\*, **Zehua Wang\***, Zhicheng Jiang\*.  
*QC-CSAA Workshop at IEEE ISVLSI 2026* (full paper). [Paper] \* Equal contribution.

## SELECTED PROJECTS

- **Open-Source Contributor, NousResearch/Hermes Agent**: Authored a **3-commit** speech-to-text provider-validation patch now in **main**, adding selected-provider and plugin-backend checks plus tests. [Commits] [Merged PR]
- **Video Real2Sim (VR2S) (MIT 6.8300)**: Co-built a single-image real-to-sim pipeline that synthesizes orbital views with a video model and reconstructs them with a pose-free 3D stack; improved perceptual fidelity and back-side appearance while characterizing a silhouette-accuracy tradeoff. [Project]
- **Fast Humanoid Loco-Manipulation via Flow Matching (MIT 6.4210)**: Compared DDPM and Flow Matching for humanoid trajectory synthesis; at five function evaluations, Flow Matching achieved **820 vs. 280 survival steps**. Enabled zero-shot loco-manipulation from walking-only data via test-time classifier guidance. [Code] [Video]
- **RL vs. SFT for Mathematical Reasoning in LLMs (MIT 6.4610)**: Implemented GMPO and its evaluation pipeline for a compute-controlled comparison of PPO, GRPO, GMPO, and RLOO with SFT on Qwen3-8B; GMPO reached **74.2%** on GSM8K versus 76.7% for SFT. [Project] [Code]

## SKILLS

**Programming:** Python, C++ — **ML/Robotics:** PyTorch, MuJoCo, Isaac Gym, Isaac Lab — **Systems:** Slurm, Docker, Linux